DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech

Branch : Electrical Engineering / Electrical Engineering (Electronics and Power)/ Electrical & Electronics Engg. / Electrical & Power

Subject Code & Name: BTEEC303 Electrical and Electronics Measurement

Semester: III

Max Marks: 60 **Date:10/02/2025** **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

		Level/CO	Marks
Q. 1	**Objective type questions. (Compulsory Question)**		**12**
1	smallest change in a measured variable to which an instrument will respond is: a) Accuracy b) Resolution c) precision d) drift		1
2	The difference between the indicated value and the true value of a quantity is : a) gross error b) dynamic error c) absolute error d) relative error		1
3	If the instrument has square law response, it can be used for the instrument of: a) DC voltage only b) AC voltage & current c) Both AC and DC quantities d) Resistance only		1
4	The primary current in CT is detected by: a) Secondary burden b) core of transformer c) load current d) none of the above		1
5	The Wheatstone bridge method of resistance measurement is ideally suitable for the Measurement a) 0.001 Ω to 1 Ω b) 0.1 Ω 100 Ω c) 100 Ω to 10 KΩ d) 100 kΩ to 10 MΩ.		1
6	Schering bridge can be used to measure which one of the followings: a) Inductance b) Resistance c) Capacitance d) Power		1
7	Which bridge measures self-inductance using a standard capacitor? a) Maxwell bridge b) Schering bridge c) Anderson bridge d) Wien bridge		1

8 A digital storage oscilloscope (DSO) is used to ___. 1

a) Generate sinusoidal signals	b) Measure instantaneous power	c) Store and analyze waveforms digitally	d) Measure electrical resistance

9 Which transducer is commonly used for temperature measurement? 1

a) RVDT	b) Thermocouple	c) LVDT	d) Strain gauge

10 Which of the following is NOT a type of transducer? 1

a) Thermistor	b) LVDT	c) Hall effect sensor	d) Oscilloscope

11 Which characteristic is crucial for selecting a transducer? 1

a) Durability	b) Sensitivity	c) Size	d) Cost

12 Signal conditioning is essential for: 1

a) Direct data transmission	b) Converting raw signals into a usable format	c) Measuring mechanical stress	d) Storing large amounts of data

Q. 2	**Solve the following.**		12
A)	Define error. Explain the different types of errors with example.	CO1	6
B)	Describe the characteristics of instruments & measurement system.	CO2	6
Q. 3	**Solve the following.**		12
A)	Draw a neat sketch of PMMC instrument and derive the equation and explain in detail.	CO1	6
B)	Explain two wattmeter method for measurement of power in three phase circuit.	CO2	6
Q. 4	**Solve Any Two of the following.**		12
A)	The inductance of moving iron ammeter is given by following expression $L=(20+10\Theta-2\Theta^2)\ \mu H$, where, Θ is deflection in radians. The spring constant is 24×10^{-6} Nm/rad. Calculate the values of deflection for a current 5A.	CO1	6
B)	With a neat, labeled diagram explain Maxwell's Inductance bridge and obtain the output equation for Lx and Rx	CO2	6
C)	With a circuit & phasor diagram derive the equation for an unknown self-inductance measurement using HAY'S bridge.	CO3	6
Q. 5	**Solve Any Two of the following.**		12
A)	Explain the working of Q Meter.	CO1	6
B)	Draw and explain Ramp type DVM in detail.	CO2	6
C)	Explain in detail the construction and working of digital storage oscilloscope.	CO3	6

Q.6	**Solve Any Two of the following.**		**12**
A)	Explain the working of strain gauge in detail.	**CO1**	**6**
B)	Explain different characteristics of transducer.	**CO2**	**6**
C)	With a neat, labeled diagram explain the operation of LVDT.	**CO3**	**6**

***** End *****